Drain Optics: Design Dossier (EXSU 620 P1)

Nov 2, 2026 · @Jehad

Technical memorandum

To: Dr. Hooshair, EXSU 620 · **From:** Project 01 group (hardware, software, calibration) · **Subject:** Drain Optics bench prototype, status at submission

After surgery, nurses judge bleeding from a drain by eye: they write down a volume and a colour word. Our bench prototype replaces the colour word with a number. A white light shines through the drain tube onto a colour sensor; blood absorbs green light much more than red, so the green-to-red ratio tells us how much blood is in the fluid. A second sensor watches the drip chamber and counts drops, which gives the flow. Multiplying the two gives the blood loss rate in millilitres per hour.

The device is a small box with a coloured light ring, a speaker and an acknowledge button. Green means normal; amber and red mean rising blood; blue with a double-single beep means no drops have fallen for 10 s, so the line may be clotted or kinked. That last alarm matters: a blocked drain looks exactly like bleeding that has stopped, and the device must not report it as good news.

What works today: the colour measurement with a four-point calibration, drop counting, the blood loss rate in mL/h, the blocked-line alarm and the light and sound alarms with a 2-minute silence. **What does not yet meet the brief:** the alarms switch on blood concentration rather than on the mL/h rate, the "rate doubles in 30 min" rule is not implemented, a fouled window and LED failure are not separately detected, and the shift total, trend and phone notification depend on the laptop service. Section D7 lists each requirement with its evidence. Everything here is a bench model with dyed water; nothing has been near a patient.

Traceability matrix

Every clinical requirement traces to the engineering requirements that serve it, the verification test that closes each one (D6), and the result at submission (D7).

Clinical	Engineering	Verification test	Result at submission
CR-1	ER-1 concentration	V-1 blind mixes	Not yet verified: calibration done (4 points, 1 series), blind mixes not run

Clinical	Engineering	Verification test	Result at submission
CR-1	ER-2 flow	V-2 weighed collection	Not yet verified: drop detection still being tuned
CR-1	ER-3 rate in mL/h	V-3 step response	Implemented; not yet tested
CR-2	ER-4 rate alarms	V-4 alarm scenarios	**Not met:** alarms use % blood thresholds, doubling rule missing
CR-3	ER-5 static line	V-5 clamp test	Implemented (10 s, blue ring); not yet tested
CR-3	ER-6 optical faults	V-6 induced faults	**Partly met:** empty tube / no light detected; lamp failure, saturation and fouled window not detected
CR-1, CR-3	ER-7 ambient light	V-7 light test	Not yet verified; no dark-frame subtraction in current firmware
CR-4	ER-8 shift record	V-8 one-hour run	**Not met on the device;** depends on the laptop service (software group)

D1 Engineering requirements

Eight engineering requirements (ER) were derived from the four clinical requirements on the project card. Each states a range, rate, latency or accuracy, has a pass/fail acceptance criterion, and names the clinical requirement it serves.

Clinical requirement (project card)	
CR-1	Blood loss reported in mL/h, not as a colour word, updated at least once a minute
CR-2	Alert when the estimated loss rate exceeds 100 mL/h, or doubles within 30 min, whichever first
CR-3	A fouled window, an empty line or a failed lamp never presents as low blood loss
CR-4	Every shift has a cumulative loss figure and a rate trend at handover and ward round

ID	Requirement	Range / rate / latency / accuracy	Acceptance criterion (pass/fail)	Serves
ER-1	Blood concentration from the green/red ratio of light through the tube	Range 0–2 % blood (dye analogue); one reading every ⩽ 0.25 s; accuracy ±0.25 % blood	3 blind mixes (0.5, 1.0, 1.5 %), 3 readings each: every 30 s mean within ±0.25 % of the true value	CR-1
ER-2	Drain flow from drop counting in the drip chamber	Range 30–600 mL/h (10–200 drops/min at 20 drops/mL); averaged over 60 s; accuracy ±10 %	At 3 clamp settings, 3 runs each: 5 min mean flow within ±10 % of the weighed collection (1 g = 1 mL)	CR-1
ER-3	Blood loss rate = % blood / 100 × flow, reported in mL/h	Output ⩾ 1 line/s; displayed value settles within 60 s of a step change	Step from clear water to a 1 % mix at fixed flow: reported rate reaches 90 % of its final value within 60 s, 3 of 3 trials	CR-1
ER-4	Clinical alarms on the estimated rate	Advisory: rate > 100 mL/h for 5 min; critical: rate doubles within 30 min, latched until acknowledged and back in range; alarm raised ⩽ 60 s after the condition is met	Replayed and live scenarios: each alarm raised within 60 s of its condition, 5 of 5; no alarm in a 30 min run below 80 mL/h	CR-2
ER-5	Static-line technical alarm (clot, kink, empty bag)	No drop for 10 s → alarm; valid for flows ⩾ 30 mL/h	Clamp shut 10 times: alarm within 15 s every time; 0 false alarms in 30 min at 30 mL/h	CR-3
ER-6	Optical fault alarms: failed lamp, empty or air-filled tube, saturation, fouled window	Detected within 5 s; reported as a technical alarm, never as 0 mL/h "normal"	Each fault induced 5 times: 20 of 20 detected and shown as a fault	CR-3

ID	Requirement	Range / rate / latency / accuracy	Acceptance criterion (pass/fail)	Serves
ER-7	Ambient light immunity of the optical head	Room lights on/off and a phone torch at 30 cm change the reading by < 0.1 % blood	5 trials of each light condition with clear water and a 1 % mix: all changes < 0.1 %	CR-1, CR-3
ER-8	Shift record: every reading logged with a timestamp; cumulative blood volume and rate trend available	Log at ⩾ 1 line/s; cumulative volume within ±15 %; trend of the last 12 h	1 h run with a known mix: logged cumulative blood within ±15 % of collected volume × known %; trend chart shows the full hour	CR-4

The accuracy and range numbers are the group's design targets, chosen so the device can resolve the 100 mL/h advisory limit; the project card sets only the clinical limits.

D2 System design

Architecture

The system has three tiers (block diagram in D4). The **sensing tier** is two optical heads on the drain line: a tube clip holding a white LED and colour sensor 1 across the tubing, and a holder clamping colour sensor 2 to the drip chamber. The **device tier** is a Raspberry Pi Pico 2 W on a Grove shield in a printed case, which reads both sensors, computes % blood, flow and blood loss rate, runs the alarms and drives the light ring, the speaker and the acknowledge button. The **service tier** is a laptop connected by USB, which logs every line to SQLite and serves the dashboard and notifications (CR-4); it is the software group's module.

The Pico works on its own: if the laptop is unplugged it keeps measuring and alarming at the bedside. If sensor 2 is missing at start-up, it falls back to a fixed flow (`FLOW_MLH`) and a frozen-reading check, and prints a warning.

Sensing approach

Concentration (sensor 1). Whole blood absorbs strongly in green and weakly in red. The Pico reads the green and red channels of the TCS34725 (1× gain, 101 ms integration) and forms g/r, averaged over 5 readings. With clear saline the ratio is the baseline, 1.3406. Each reading becomes an absorbance-like value and is converted to % blood by linear interpolation between the calibration points (D7):

> **A = 1 − (g/r) / 1.3406** → % blood from (A, %) = (0, 0), (0.64, 0.5), (0.78, 1.0), (0.86, 2.0)

Readings below 0.3 % are set to 0 (noise deadband).

Flow (sensor 2). Sensor 2 keeps its own white LED on and runs at 2.4 ms integration. It is polled about every 3 ms while sensor 1 integrates. A falling drop changes its clear reading by more than 4 % from a slowly-tracking baseline; a 150 ms lockout makes one drop count once. Drops in the last 60 s give drops/min, and:

> **Flow (mL/h) = drops/min × 60 / drop factor** (drop factor assumed 20 drops/mL; not printed on our IV set)

> **Blood loss rate (mL/h) = % blood / 100 × flow**

Alarm rules

State	Rule as required (card)	Rule as implemented	Light	Sound
Normal	Rate below limits	% blood < 0.5	Green, ring fills with % blood	Silent
Advisory	Rate > 100 mL/h for 5 min; back below for 10 min to clear	% blood ⩾ 0.5; clears below 0.4	Amber	1 slow beep/s, 500 Hz (card: 3 pulses)
Critical	Rate doubles within 30 min; latched until acknowledged and back in range	% blood ⩾ 1.8; steps down below 1.7	Red	3 fast beeps/s, 1000 Hz (card: 10 pulses)
Technical: static line	No flow	No drop for 10 s	Blue (card: cyan)	Beep-beep … beep, 750 Hz
Technical: no light	Lamp failure, empty line	Sensor 1 clear channel < 10 counts	Purple	Long beep every 2 s

The acknowledge button silences the sound for 2 min; the light stays on, and the sound returns at once if a new or worse alarm starts. The gap between the required and implemented rules is reported in D7 (ER-4).

Data model

The Pico prints one line per reading (about 5 per second) over USB serial:

```
BLOOD_PCT:0.85% | DROPS/S:0.50 | DROPS/MIN:30 | FLOW:90.0 mL/h |
BLOOD_RATE:0.77 mL/h | STATE:ADVISORY | DROPS:412 | SILENCED:False
```

The laptop service is designed to store each line with its arrival time in one SQLite table:

Column	Type	Meaning
t	REAL	Unix time the line arrived (s)
blood_pct	REAL	% blood
drops_min	REAL	Drops per minute, last 60 s
flow_mlh	REAL	Drain flow (mL/h)
rate_mlh	REAL	Blood loss rate (mL/h)
state	TEXT	NORMAL, ADVISORY, CRITICAL, CHECK_DRAIN, AIR_BLANK
drops_total	INTEGER	Drops since start
silenced	INTEGER	1 if the sound is silenced

Cumulative shift loss = Σ rate × Δt over the shift. No patient identifiers are stored.

Alternatives rejected

Alternative	Why rejected
One sensor only, with a flow typed in by hand	Colour alone cannot give mL/h: a slow very bloody drain and a fast lightly bloody one look the same. A typed flow goes stale when the clamp moves, and a frozen colour reading is a weak sign of a clot. Built as `monitor_1sensor.py`, replaced by the drop counter.
"Δabsorbance per minute" from one sensor as a rate	Measures how fast the colour changes, not how much fluid flows; gives 0 for steady bleeding.

Alternative	Why rejected
Pocket scale under the bag as the live flow channel	The card forbids the reference instrument as a live channel; the scale is used only for dilutions and flow checks.
Cuvette holder (sample taken off the line)	Needs manual sampling, so no continuous reading; replaced by the flow-through tube clip.
NeoPixel pixel as the light source	The ring is the alarm indicator; using it as the lamp ties the two together. A separate white LED on GP15 is used.
CircuitPython firmware with laptop-side rules (`code.py`)	Written first, with LED modulation, dark frames and full fault checks. The team's bench work moved to MicroPython; porting those checks back is listed in D7.
I2C multiplexer for two sensors at address 0x29	Not in the kit; a second bus (software I2C on GP6/GP7) does the job.

D3 Component, module and system chart

Modules

Module	Components	Function	Owner
M1 Concentration head	Tube clip v4 (black PLA), white LED + 330 Ω, TCS34725 sensor 1	Shines light through the tubing and measures green and red	Hardware
M2 Drop counter head	Drip holder v3 (black PLA), TCS34725 sensor 2 (own LED on)	Detects each falling drop in the drip chamber	Hardware
M3 Controller	Pico 2 W, Grove shield, half-size breadboard, `monitor_2sensor.py`	Reads sensors, computes % blood, flow and rate, runs alarms, prints data	Hardware
M4 Indicator and alarm	NeoPixel Ring 16 under translucent cap, PAM8302 + 8 Ω speaker, panel pushbutton	Shows state by colour, sounds tone patterns, takes acknowledgement	Hardware

Module	Components	Function	Owner
M5 Enclosure	Case v3 base, snap lid, light cap (printed)	Holds M3 and M4; button and grille on the front, cable slots	Hardware
M6 Laptop service	Python, pyserial, SQLite, dashboard, messaging	Logs, cumulative loss, trend, notifications (CR-4)	Software
M7 Bench rig	IV bag with dyed water, giving set with drip chamber, roller clamp, tubing, cup	Phantom drain line	Calibration
M8 Reference	Pocket scale 0.1 g, syringes, measuring cup	Dilutions and flow checks only, never live	Calibration

Interfaces

ID	From → To	Signal	Range
IF-1	Sensor 1 → Pico	I2C0, GP8 SDA / GP9 SCL, 100 kHz, address 0x29	3.3 V logic; R, G, B, C counts 0–65 535 per 101 ms
IF-2	Sensor 2 → Pico	Software I2C, GP6 SDA / GP7 SCL, 400 kHz, address 0x29	3.3 V logic; clear counts 0–65 535 per 2.4 ms, polled every ~3 ms
IF-3	Pico → white LED	GP15 digital, low = on (VBUS → 330 Ω → LED → GP15)	0 / 3.3 V; about 3–6 mA
IF-4	Light, LED → sensor 1	Through ×4.4 tube channel and ×3.5 window, beam 12.2 mm above clip base	Visible white light
IF-5	Pico → NeoPixel ring	GP18 single-wire data, 800 kHz; 5 V supply from VBUS	16 × RGB, each 0–255 (used ⩽ 30)
IF-6	Pico → PAM8302	GP20 PWM tone into A+; A− to GND	500–1000 Hz, duty 0–65 535 (set 100)
IF-7	PAM8302 → speaker	Bridged audio output	Up to 2.5 W into 8 Ω at 5 V
IF-8	Button → Pico	GP16 input, internal pull-up	1 = released, 0 = pressed
IF-9	Pico → laptop	USB CDC serial text, one line per reading	~5 lines/s; fields as in D2 data model

ID	From → To	Signal	Range
IF-10	Laptop → Pico	USB VBUS power	5 V, ⩽ 500 mA
IF-11	Fluid, drip chamber → tube clip	Gravity flow through IV tubing (~4.1 mm OD)	0–600 mL/h on the bench
IF-12	Laptop → phone	Messaging API (planned)	Text alert, no patient identifiers

D4 System diagram set

Block diagram

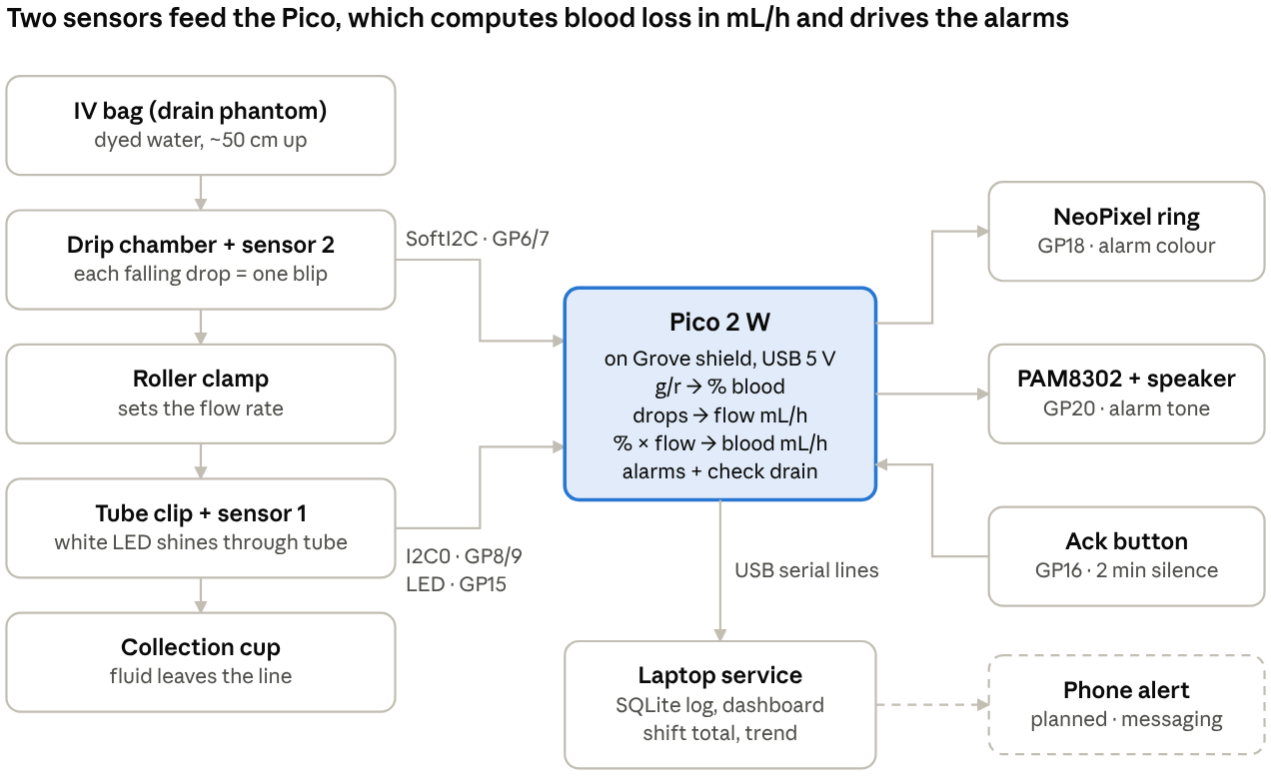


Figure 1. Block diagram

Fluid runs down the left column. Sensor 2 reads the drip chamber and sensor 1 reads the tube; each has its own I2C bus because both chips have the fixed address 0x29. Dashed parts are planned.

Wiring diagram

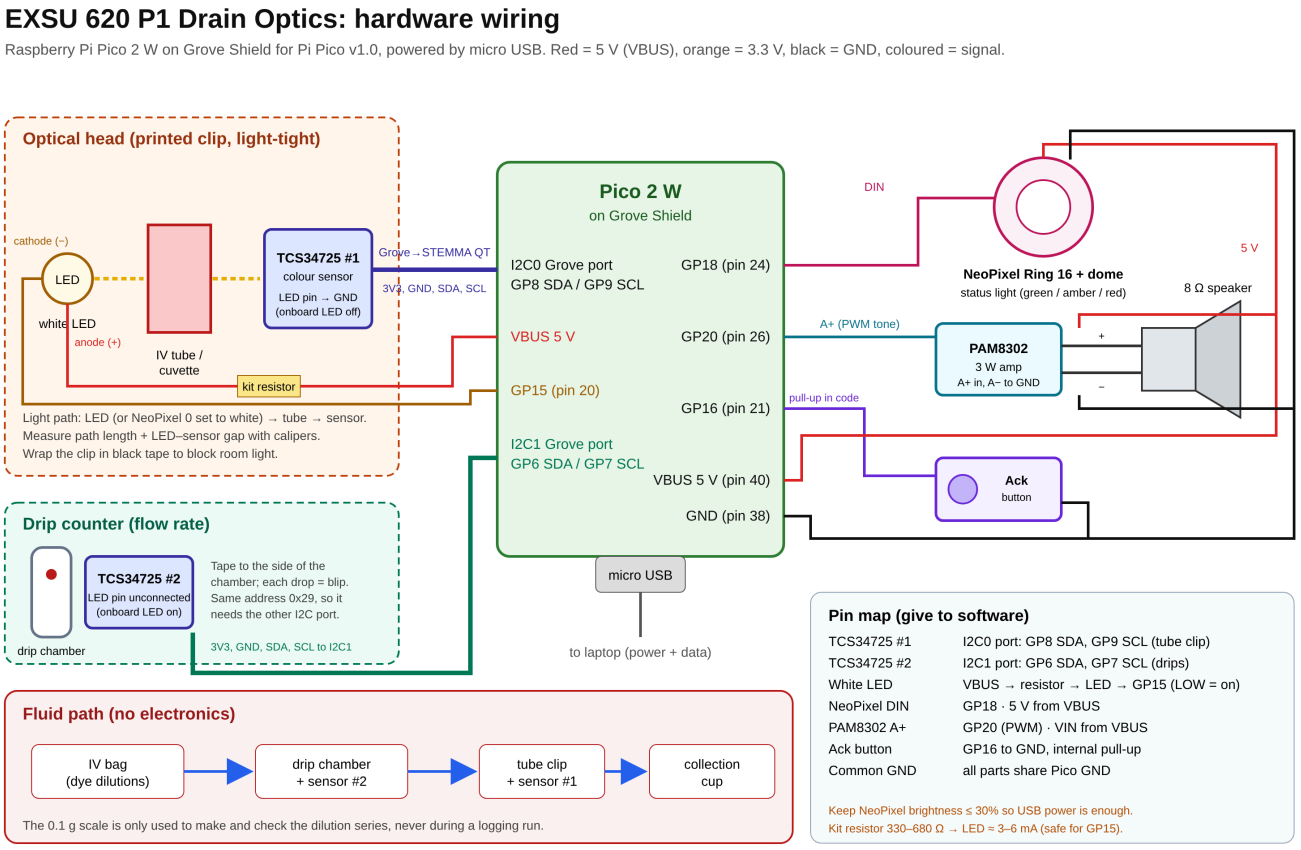
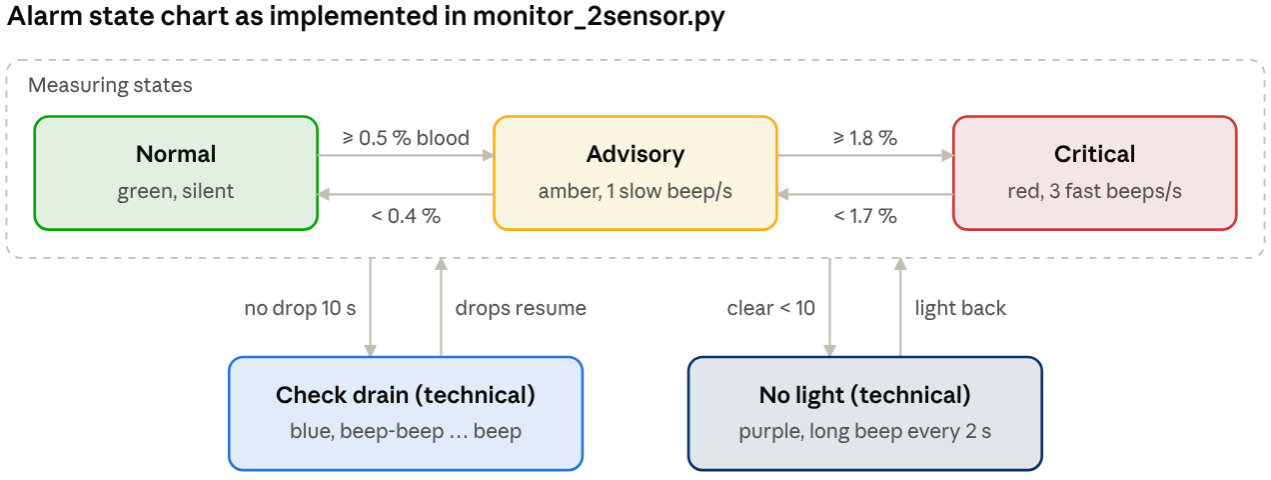


Figure 2. Wiring diagram and pin map. Sensor 2 is wired by four jumpers to Pico pins 8/9/10 (GND, GP6, GP7) and 3V3 (pin 36).

Alarm state chart



Ack button silences sound for 2 min in any alarm state; the light stays on. Card requires mL/h rules (D7, ER-4).

Figure 3. Alarm state chart

Sequence diagram

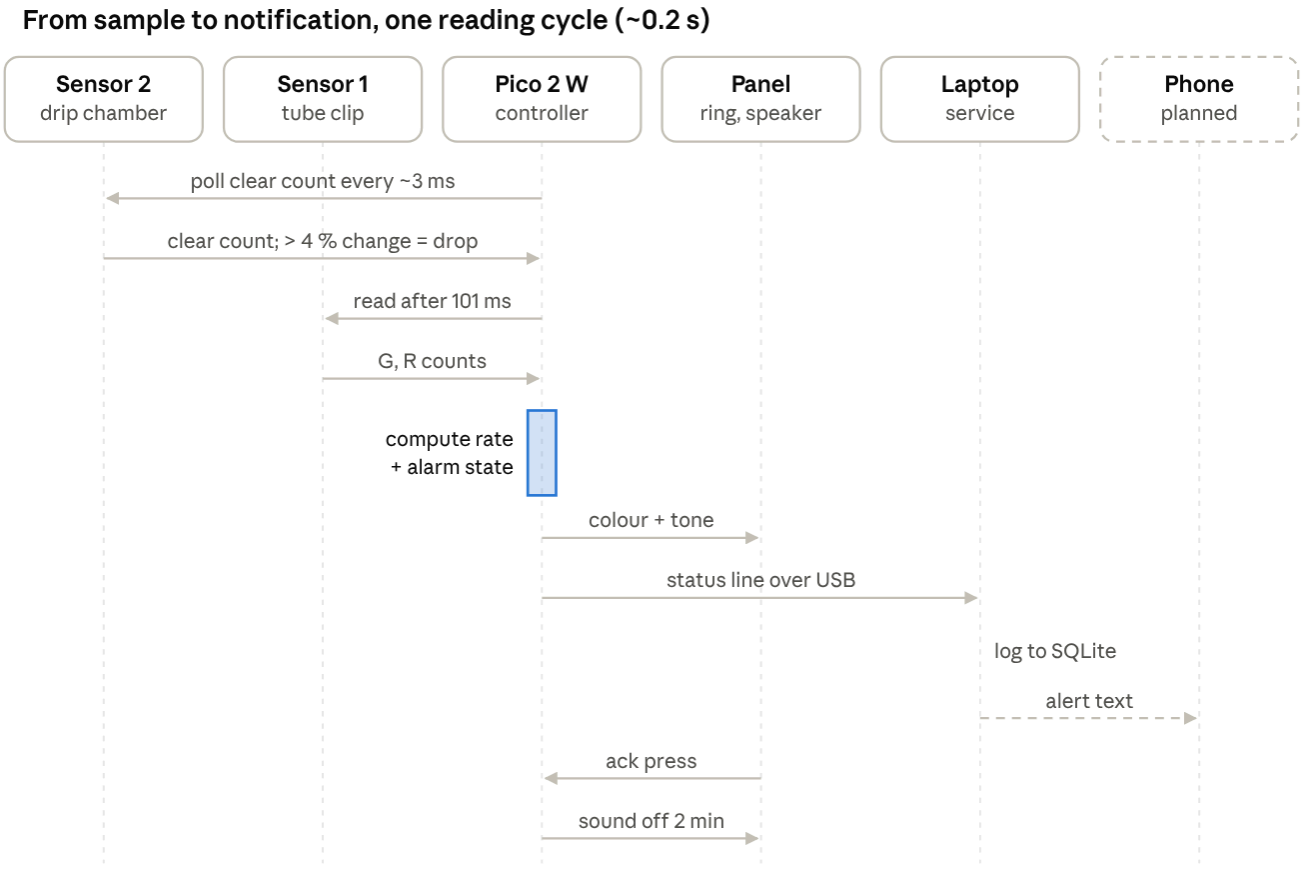


Figure 4. Sequence diagram, sample to notification

Sensor 2 is polled continuously while sensor 1 integrates, so no drop is missed. The phone message (dashed) is planned for the laptop service and is not yet implemented.

D5 Implementation, deployment and documentation

Firmware

`monitor_2sensor.py` (MicroPython on the Pico 2 W) is the current firmware. It keeps the team's calibrated sensor 1 settings and points, adds the drop counter, the blood loss rate and the check-drain alarm, and silences for 2 min on the button. Drop detection thresholds are still being tuned on the bench.

Setting	Value	Meaning
`BASE_RATIO`	1.3406	g/r with clear saline (0 % blood)
`CAL_POINTS`	(0, 0), (0.64, 0.5), (0.78, 1.0), (0.86, 2.0)	(absorbance A, % blood)

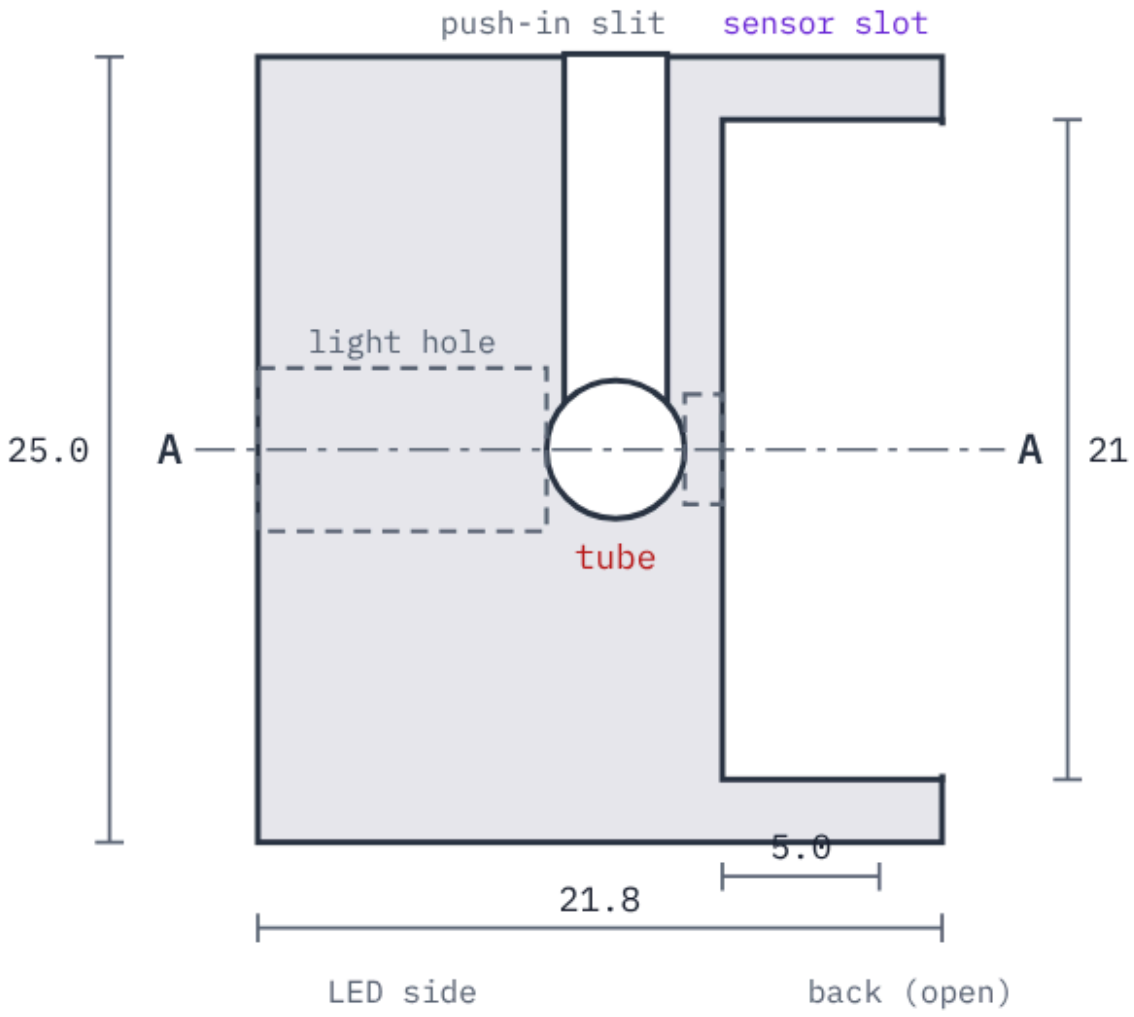
Setting	Value	Meaning
`DROP_FACTOR`	20 drops/mL	Assumed standard adult IV set
`DROP_THRESHOLD`, `LOCKOUT_MS`	4 %, 150 ms	Drop detection
`FLOW_WINDOW_MS`	60 s	Flow averaging window
`STATIC_MS`	10 s	No drop this long = check drain
`ADVISORY_PCT`, `CRITICAL_PCT`, `HYST_PCT`	0.5 %, 1.8 %, 0.1 %	Alarm thresholds (provisional)
`SILENCE_MS`	120 s	Silence after an ack press
`VOLUME`	100 / 65 535	PWM duty for the tone
`FLOW_MLH`	180 mL/h	Fallback flow if sensor 2 is missing

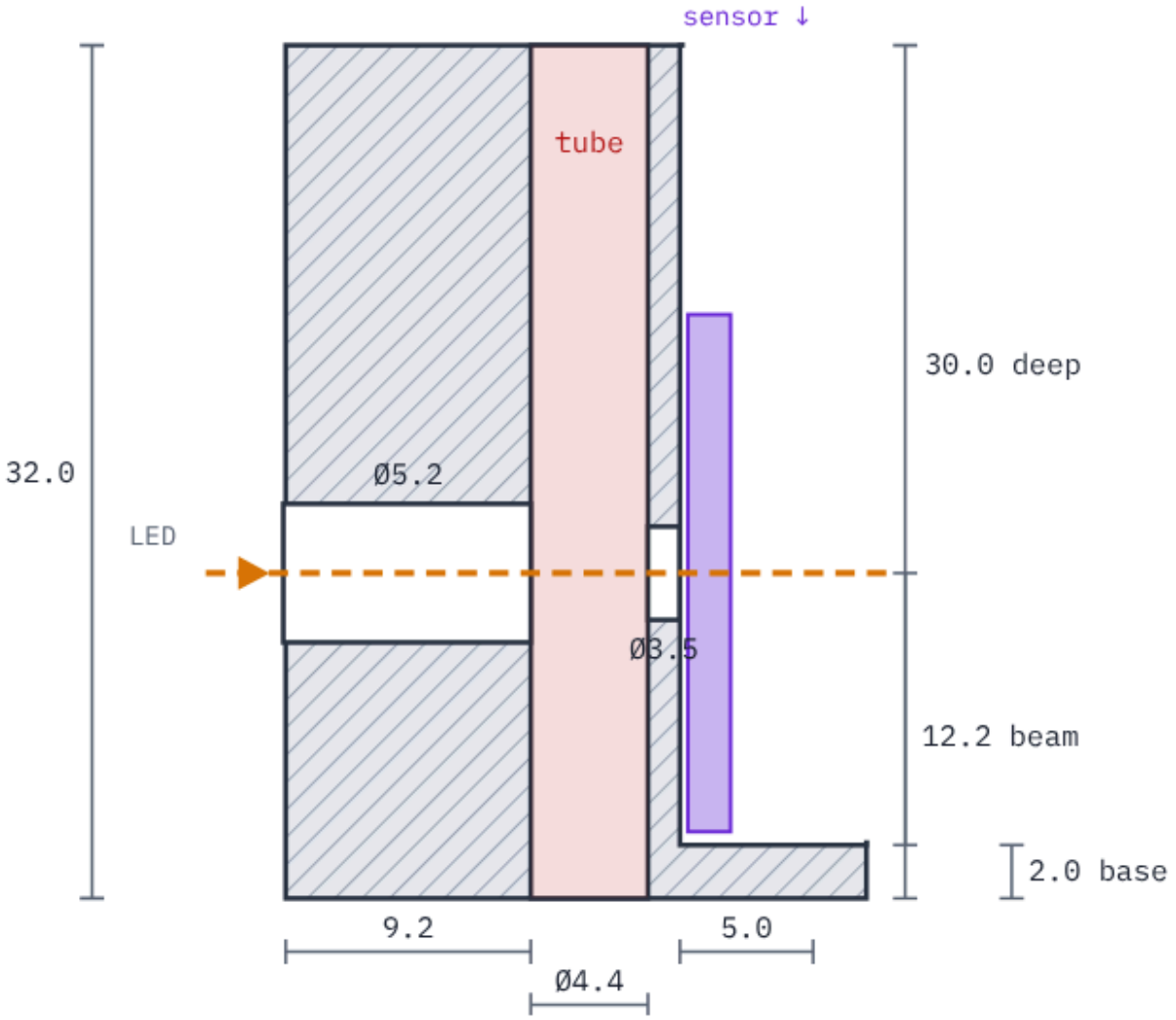
Printed parts

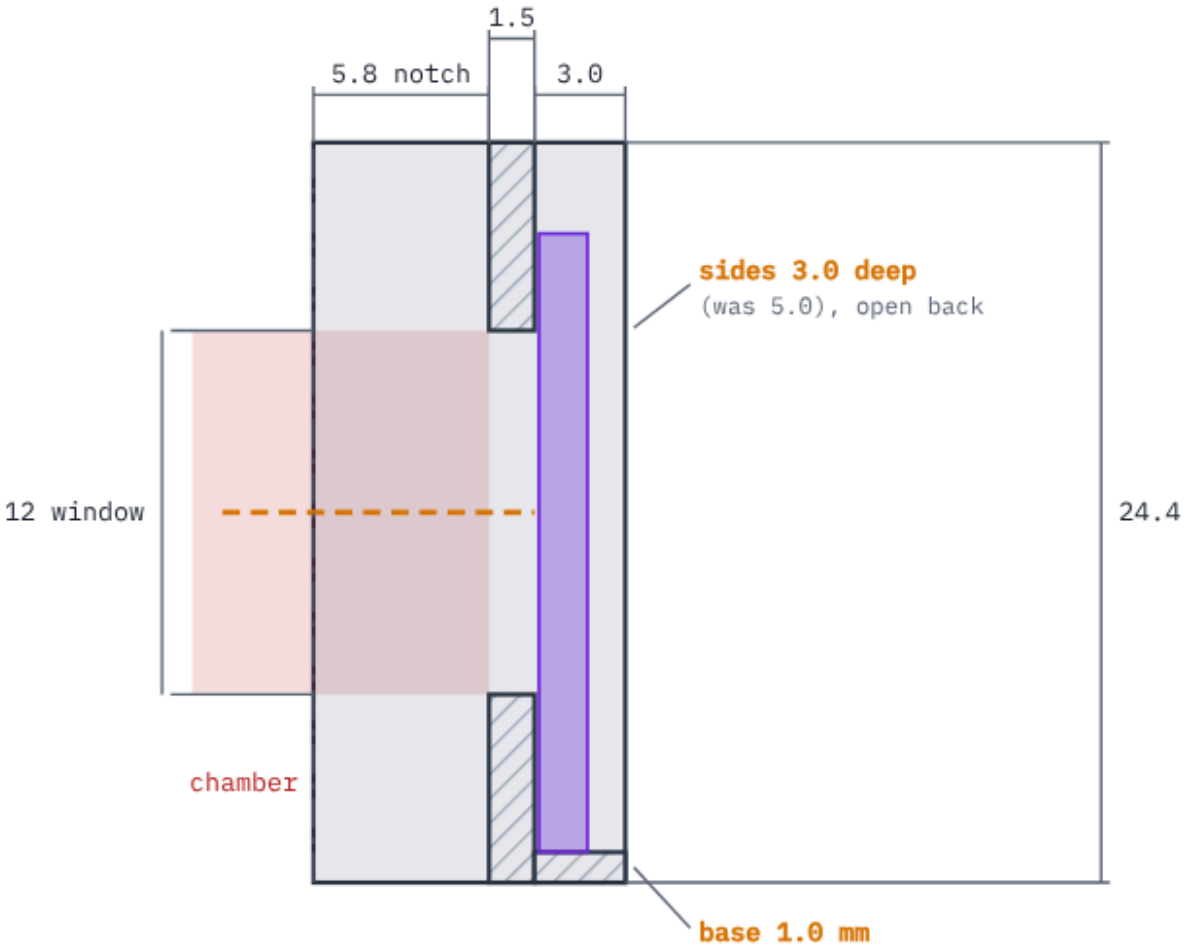
Parts were designed in Python CAD scripts and exported as STL and STEP. Optical parts are black PLA so room light cannot leak through the walls.

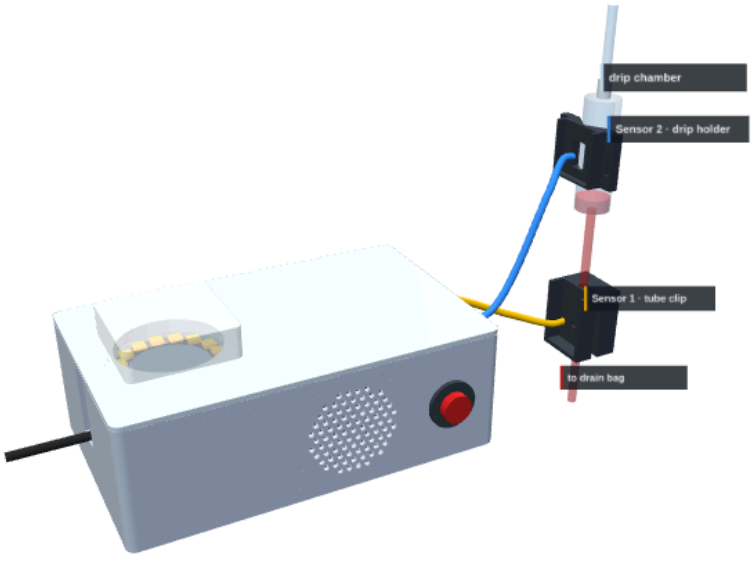
Part	Version	Size (mm)	Key features	Print settings
Tube clip (sensor 1)	v4	21.8 × 25 × 32	×4.4 tube channel with push-in slit, ×5.2 LED hole, ×3.5 window, open-back sensor slot 5.0 wide on a 2.0 base, beam at 12.2	Black PLA, flat on the 21.8 × 25 face, no supports, 0.2 mm
Drip holder (sensor 2)	v3	10.3 × 37 × 24.4	120° V-notch fits chambers ×8–22, 12 mm window, strap ears, 1.0 base	Black PLA, upright, no supports, 0.2 mm
Case	v3	Base 152 × 94 × 50; lid 6.6; light cap 44 × 44 × 15.3	×16.2 button hole, ×36 speaker grille, ring under ×41 window, cable slots	White/grey PLA, 3 walls, 20 %; cap translucent, 2 walls, no infill

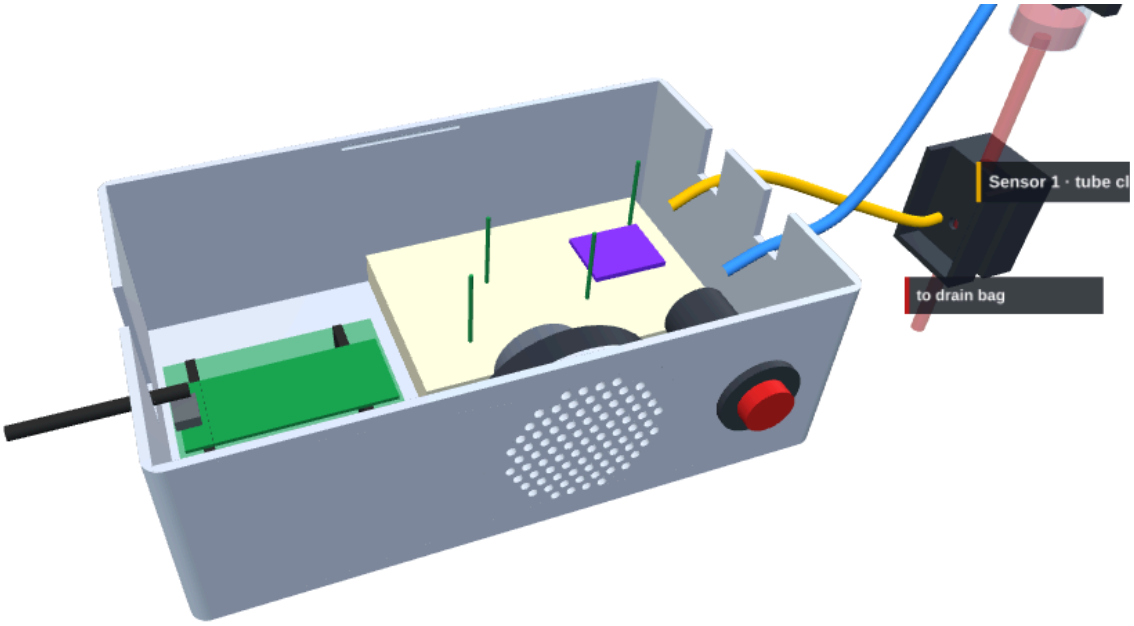
The tube clip went through four versions after test fits: v2 widened the sensor slot (3.4 → 5.0) and window (3.0 → 3.5); v3 opened the back so the sensor board fits; v4 added a base and centred the beam on the sensor chip.




sensor ↓
tube
30.0 deep
32.0
Ø5.2
LED
Ø3.5
12.2 beam
2.0 base
9.2
5.0
Ø4.4


1.5
5.8 notch
3.0
sides 3.0 deep
(was 5.0), open back
12 window
24.4
chamber
base 1.0 mm


drip chamber
Sensor 2 · drip holder
Sensor 1 · tube clip
to drain bag



Figures 5–9. Printed parts (dimensions in mm) and case CAD renders.

Deployment instructions

1. Flash MicroPython for the Pico 2 W (hold BOOTSEL, plug in, drop the .uf2 on the drive).
2. Wire as in Figure 2: sensor 1 to Grove I2C0, sensor 2 to GP6/GP7 + 3V3 + GND, white LED to GP15 through 330 Ω from VBUS, ring to GP18, amplifier A+ to GP20, button to GP16 and GND.
3. Clip the tube into the tube clip and strap the drip holder to the drip chamber at the height where drops fall. Wrap both in black tape; nothing may wobble.
4. Open `monitor_2sensor.py` in Thonny, check `DROP_FACTOR` against the IV set packet, save it to the Pico as `main.py` so it runs at power-up.
5. Check the start-up line says "2 sensors". If it warns that sensor 2 is missing, check its wiring.
6. Optionally connect the laptop service to the Pico's serial port to log and show the dashboard.
7. Recalibrate (D7) if the clip, tubing, LED, resistor or IV set changes.

D6 Verification plan

All tests run on the bench rig with the device in its final position, lid on, in normal room light, after the gain, LED, clip and IV set are frozen. Mixes are made by weight on the 0.1 g scale (1 g ≈ 1 mL). The scale is never used during a logging run.

Test	Closes	Method	Apparatus	Sample size	Acceptance criterion	Sign-off
V-1	ER-1	Blind mixes prepared by a second person, read in shuffled order after a 1 min flush	Rig, 0.1 g scale, dye stock, syringes	3 mixes × 3 readings of 30 s	Every mean within ±0.25 % blood	Calibration lead
V-2	ER-2	Timed collection vs counted flow	Rig, cup, 0.1 g scale, stopwatch	3 clamp settings × 3 runs of 5 min	Mean flow within ±10 % of weighed volume × 12	Calibration lead
V-3	ER-3	Switch the bag from clear water to a 1 % mix at fixed flow; time the reported rate	Rig, serial log	3 steps	90 % of final value within 60 s	Hardware lead
V-4	ER-4	Feed recorded rate files (steady 50, 120 mL/h for 6 min, a doubling) to the rules, then live	Laptop, recorded logs, rig	5 runs per alarm + 30 min below 80 mL/h	Alarm within 60 s of condition, 5/5; no false alarm	Software lead
V-5	ER-5	Shut the clamp while running	Rig, stopwatch	10 closures + 30 min at 30 mL/h	Alarm within 15 s, 10/10; 0 false alarms	Hardware lead
V-6	ER-6	Induce faults: unplug LED, drain tube (air), torch into clip (saturation), smear the window	Rig, phone torch, grease pen	4 faults × 5	20/20 shown as technical alarm within 5 s	Hardware lead
V-7	ER-7	Room lights on/off and torch at 30 cm with clear water and a 1 % mix	Rig, phone torch	5 per condition	Change < 0.1 % blood	Calibration lead
V-8	ER-8	1 h unattended run with a known mix; compare logged cumulative blood with collected volume × known %	Rig, laptop service, cup, scale	1 run of 1 h	Within ±15 %; log complete; trend shows the hour	Software lead

D7 Verification report

At submission one requirement has been partly met and two are not met; the other five are implemented or calibrated but their verification tests have not yet been run. Each result below names the requirement it closes. Blank result cells are tests still to run.

Calibration curve

One calibration series was measured with clear saline and three dye mixes (dilutions by volume):

Mix (% blood)	Absorbance A	g/r (= 1.3406 × (1 − A))
0 (clear saline)	0.00	1.3406 (measured)
0.5	0.64	0.483
1.0	0.78	0.295
2.0	0.86	0.188

The curve is steep at low concentration and flattens above 1 %: the sensor separates 0 from 0.5 % clearly but has less resolution between 1 and 2 %. Because the firmware interpolates straight through these four points, residuals at the points are zero by construction; real residuals need the independent blind mixes of V-1. Readings under 0.3 % are set to 0. Above 2 % the curve is extrapolated, so results there are not defensible. Drift over time has not yet been measured (to log during V-8).

Pass or fail per requirement

Req.	Result	Evidence	Cause / next action
ER-1	Not yet verified	4-point calibration above	Run V-1; repeat the series 3 times
ER-2	Not yet verified	Sensor 2 detected on its bus; drop counting runs; detection still being tuned	Tune threshold, run V-2; confirm drop factor (20 assumed, not on packet)
ER-3	Implemented, not yet verified	Firmware prints rate = % × flow every reading	Run V-3

Req.	Result	Evidence	Cause / next action
ER-4	**Not met**	Firmware alarms on % blood (0.5 %, 1.8 %), not on mL/h; no 5-min persistence; doubling rule absent; amber/red tone patterns differ from 3 / 10 pulses	Drop counter only just added; move thresholds to rate and add rules (laptop or Pico)
ER-5	Implemented, not yet verified	No drop for 10 s gives blue ring and beep pattern	Run V-5; valid only above ~30 mL/h; card colour is cyan
ER-6	**Partly met**	Empty tube / no light detected (clear < 10)	Lamp failure only caught if fully dark; saturation and fouled-window checks exist in `code.py` but not in the MicroPython firmware: port them
ER-7	Not yet verified	Black clip and tape; no dark-frame subtraction	Run V-7; add LED-off dark reading if it fails
ER-8	**Not met** on the device	Pico prints every reading; logging, total and trend are in the laptop service	Software group to complete and run V-8

Fault dataset, latency, sensitivity and specificity

The labelled fault dataset (V-6), alarm latency (V-3, V-4, V-5) and sensitivity/specificity of the alarms are to be recorded from the tests above. Design latencies are: rate update every reading (~0.2 s) with a 60 s flow window, check-drain after 10 s without drops, no-light on the next reading.

D8 Failure modes and mitigation

Ratings 1 (low) to 5 (high): S = severity, O = occurrence, D = difficulty of detection; RPN = S × O × D.

ID	Failure mode	Effect	S	O	D	RPN	Mitigation	Status
FM-1	Drain line blocked (clot, kink or clamp shut)	Flow stops; looks like bleeding has stopped	5	3	2	30	Check-drain technical alarm: blue ring + beep pattern when no drops for 10 s (or reading frozen 60 s with one sensor)	Implemented; test pending
FM-2	White LED fails or its lead comes loose	Less light reaches the sensor; can read as wrong blood level	5	2	3	30	No-light alarm when sensor 1 clear channel < 10 counts	Partly: only total darkness caught
FM-3	Optical window fouled (film, dried blood, condensation)	Reads redder or darker than true	4	3	4	48	g/r ratio cancels part of it; add red-reference window check	Not implemented
FM-4	Air bubbles or empty tube in the clip	Jumpy or no valid reading	4	3	2	24	No-light alarm for an empty tube; 5-reading average smooths bubbles	Implemented
FM-5	Room light or a torch leaks into the clip	False concentration	3	3	3	27	Black PLA clip wrapped in black tape; dark reading between samples planned	Partly
FM-6	Sensor saturates (bright light or gain too high)	Ratio meaningless, may read as clear	4	2	3	24	Gain fixed at 1x; add saturation check (any channel ≥ 60 000)	Not implemented
FM-7	Tube moves or is not seated in the clip	Light path changes; wrong % blood	4	3	3	36	Push-in slit holds the tube; clip taped; recalibrate after any change	Implemented

ID	Failure mode	Effect	S	O	D	RPN	Mitigation	Status
FM-8	Blood concentration above the calibrated range (> 2 %)	Extrapolated reading, poor accuracy	4	2	3	24	Extend the dilution series; flag readings above 2 %	Not implemented
FM-9	Wrong flow value (drop factor wrong or entered flow out of date)	Blood loss rate wrong by a fixed ratio	4	3	4	48	Check flow by timed 5-min collection; set DROP_FACTOR / FLOW_MLH	Not verified
FM-10	Sensor cable unplugged or I2C read error	Script stops; reading freezes	5	2	2	20	Catch read errors and switch to a technical alarm; laptop sees lines stop	Not implemented
FM-11	Pico crashes or loses USB power	Device silent; no alarm at all	5	2	3	30	Script saved as main.py so it restarts at power-up; watchdog timer planned	Partly
FM-12	Speaker or amplifier fails, or volume too low	Alarm not heard	4	2	3	24	Light ring alarms at the same time; VOLUME and amp screw adjustable; start-up beep test planned	Partly
FM-13	NeoPixel ring fails or is too dim	Visual alarm missed	3	2	3	18	Speaker alarms at the same time; ring under translucent cap	Implemented
FM-14	Alarm silenced and forgotten	Missed escalation	4	3	2	24	Silence lasts only 2 min; a new or worse alarm sounds at once	Implemented
FM-15	Ack button stuck down or shorted	Sound stays silenced	4	1	3	12	Require button release between presses; show silenced state on the ring	Not implemented

Highest risks are a fouled window and a wrong flow value (RPN 48), then a loose tube in the clip (RPN 36); these are next on the work list.

Appendix

Course constraints

- **Bench only:** the rig is a dyed-water phantom; nothing goes near a patient.
- **No identifiers:** no patient information is stored or sent.
- **Measurement integrity:** the 0.1 g scale is used only to make and check mixes and flows, never live.
- **Parts:** all parts are from the kit. The only change from the card is MicroPython instead of CircuitPython, which the team used for faster bench iteration.

Files (project `hardware/` folder)

What	File
Current firmware	`monitor_2sensor.py`
Earlier firmware	`monitor_1sensor.py`, `monitor.py`, `firmware/code.py` (CircuitPython, with fault checks)
Bench test scripts	`sensor1_read.py`, `drip_test.py`, `drip_signal.py`, `hw_test.py`
Wiring	`wiring-diagram.png`, `full-wiring.png`
Tube clip v4	`print/DrainOptics_TubeClip_v4.stl` / `.step`, `tube_clip_v4_cad.py`
Drip holder v3	`print/DrainOptics_DripHolder_v3.stl` / `.step`, `drip_holder_v3_cad.py`
Case v3	`print/T6_caseA_base_v1`, `T6_caseA_lid_v1`, `T6_light_cap_v1`, `enclosure/enclosure_v3_cad.py`